Politeknik
Negeri
Batam

Deep Learning - Transfer Learning

Week 13
D4 Robotics Engineering Study Program
Department of Electrical Engineering
Politeknik Negeri Batam

Machine Learning

Traditional Deep Learning Challenges

Permasalahan Utama:

- Membutuhkan dataset yang sangat besar (jutaan sampel)
- Waktu training yang sangat lama
- Komputasi yang mahal (GPU high-end)
- Overfitting pada dataset kecil

ImageNet
14 juta gambar

Training Time
2-4 minggu

GPU Hours
1000+ jam

Cost
$10,000+

Solusi: Transfer Learning memungkinkan kita memanfaatkan model yang sudah dilatih untuk mengatasi keterbatasan ini.

Transfer Learning

Transfer Learning:

Metode dalam machine learning yang memanfaatkan pengetahuan yang diperoleh dari satu tugas (source domain) untuk meningkatkan pembelajaran pada tugas yang berbeda namun terkait (target domain).

Konsep Dasar:

Source Domain → Knowledge Transfer → Target Domain

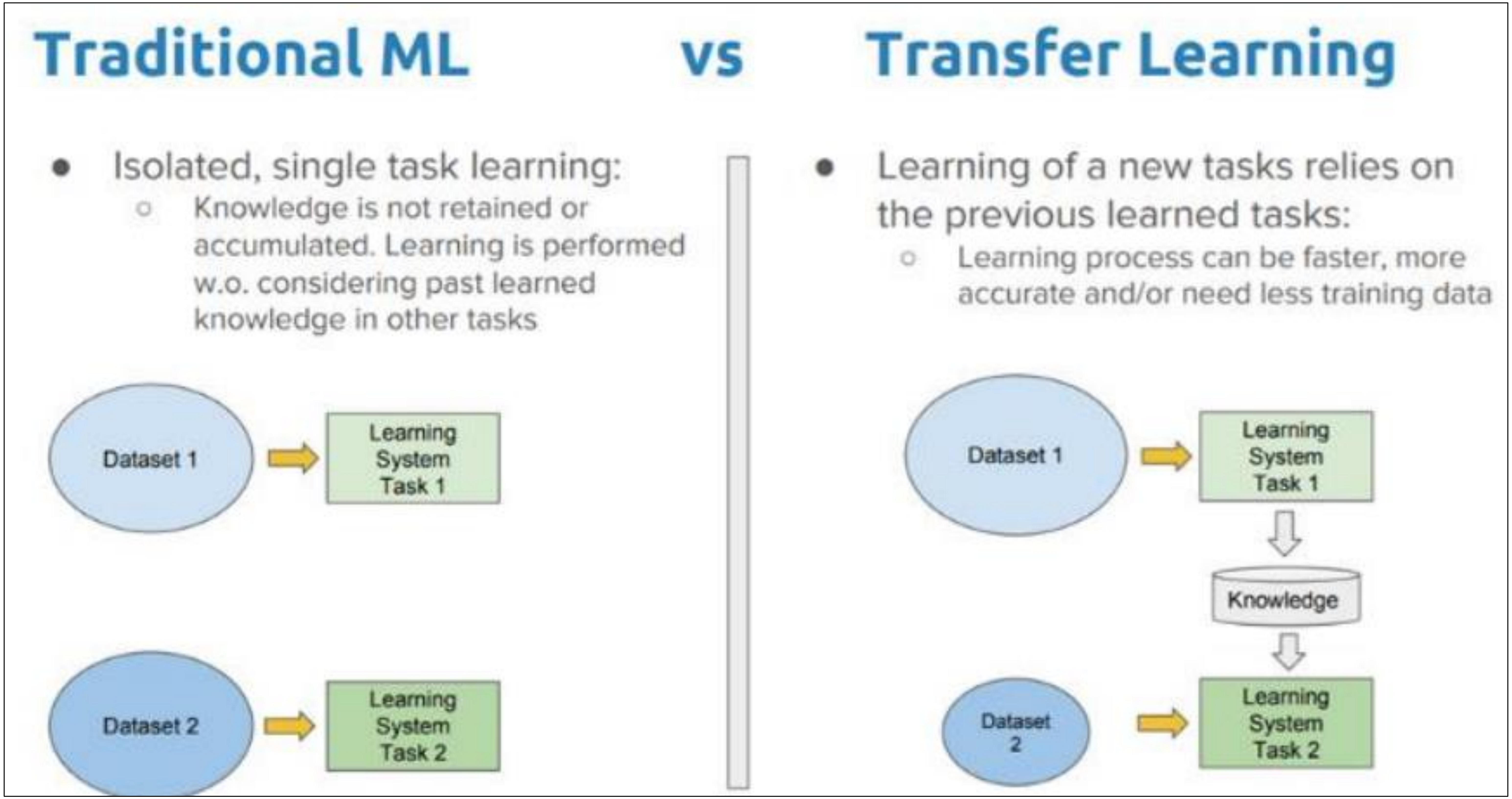
Source Domain

- Dataset besar (ImageNet)
- Task umum (klasifikasi objek)
- Model pre-trained
- Feature representations

Target Domain

- Dataset spesifik (medical images)
- Task khusus (diagnosis)
- Model yang disesuaikan
- Domain-specific features

Transfer Learning Illustration

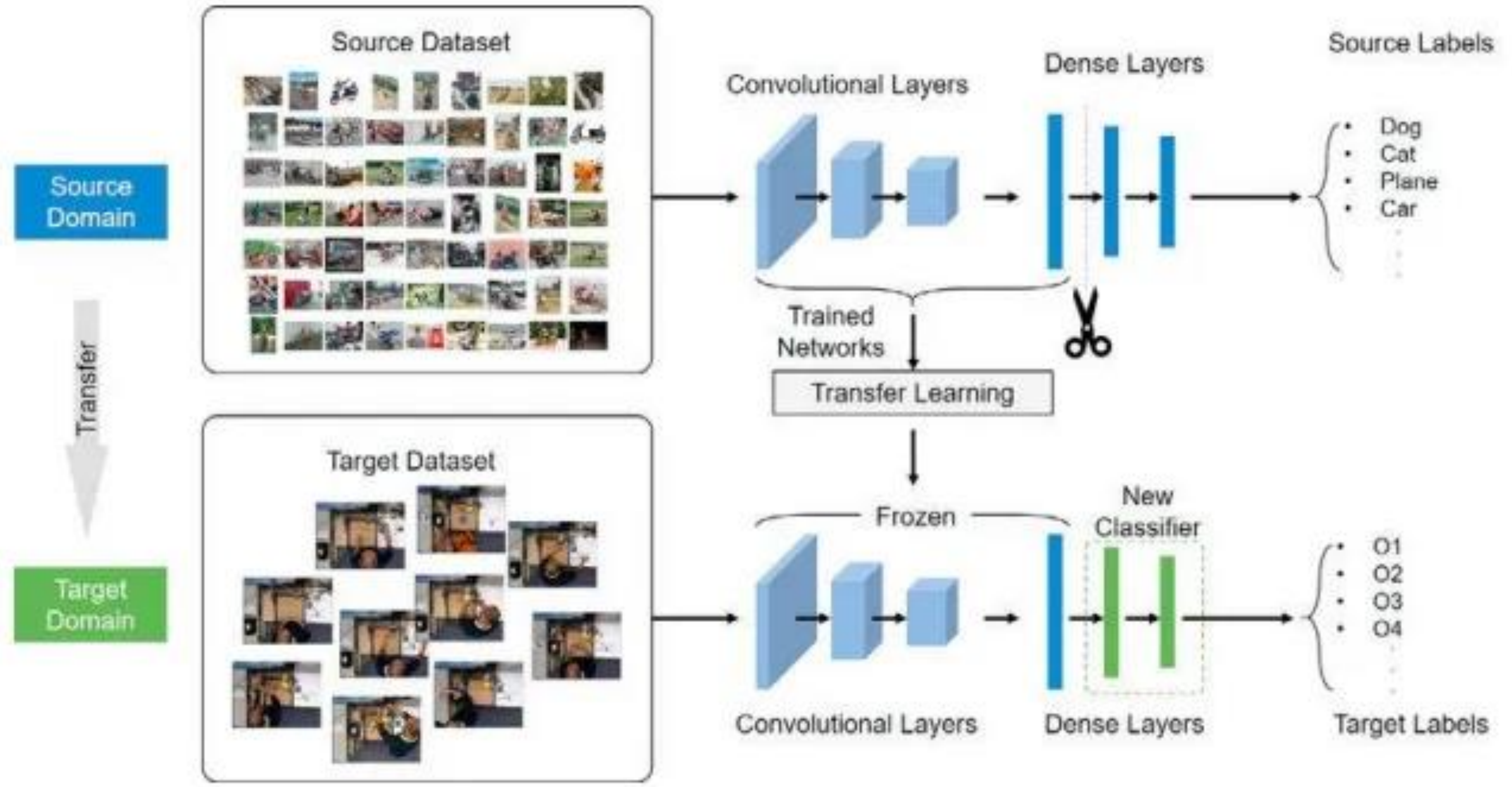


Types of Transfer Learning

Metode	Karakteristik	Datset Size
Fixed Feature Extractor	Freeze pre-trained layers	Kecil (<1K)
Fine-tuning	Update semua layers	Sedang (1K-10K)
Pre-trained as Initialization	Starting point training	Besar (>10K)

Fixed Feature Extractor

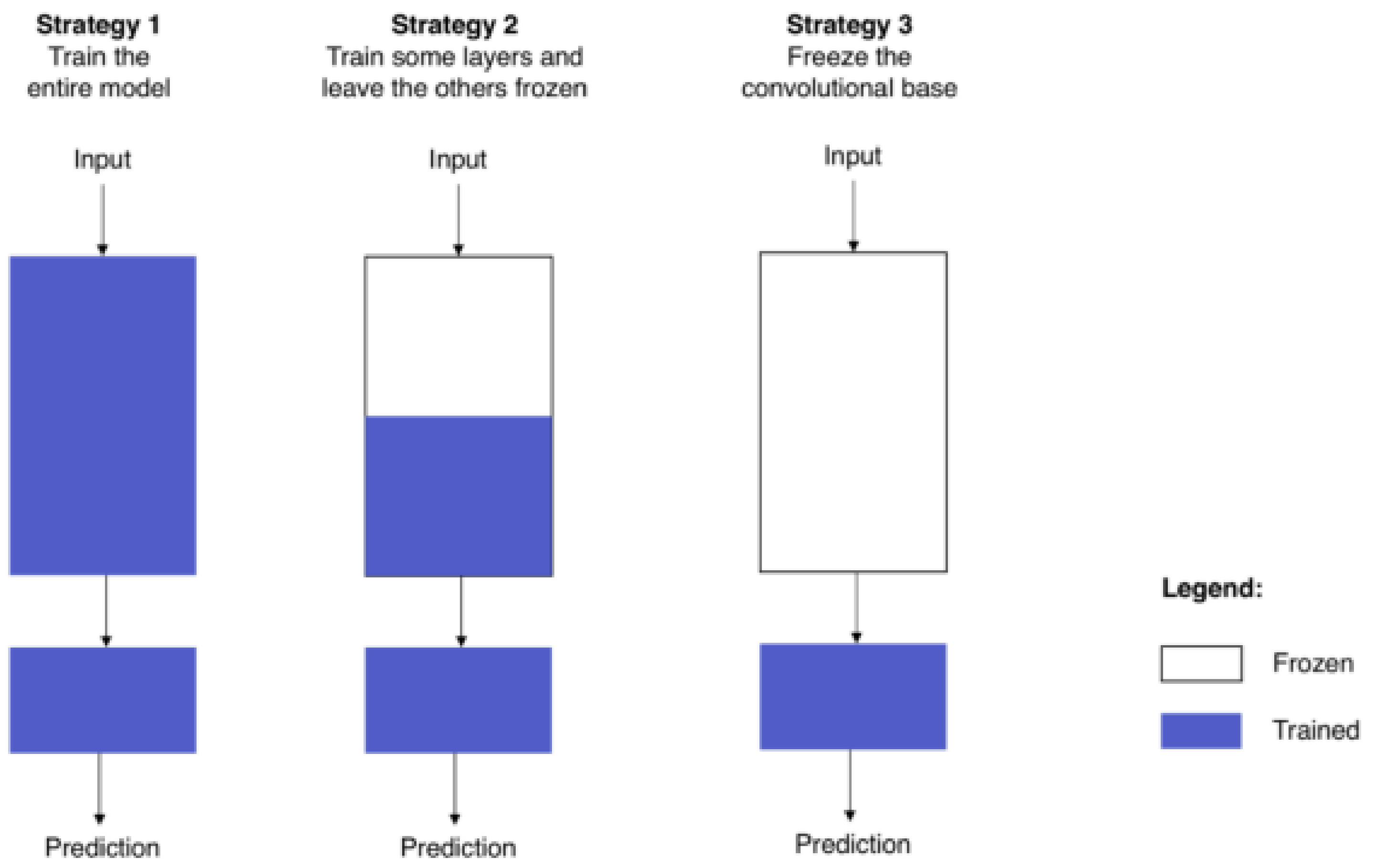
Arsitektur model yang lama (Source Domain) akan digunakan sebagai feature extractor, kemudian bagian classifier (bagian akhir neural networks) dimodifikasi.



Fine-Tuning

Metode ini mirip dengan teknik Fixed Feature Extractor, hanya saja pada saat proses pelatihan dengan menggunakan target domain, **frozen** tidak digunakan

Using Pre-Trained Model



Metode ini memanfaatkan checkpoints dari pelatihan model yang lain

Applications in Computer Vision

An Effect of Transfer Learning and Data Augmentation for Fruit And Vegetable Type Classification

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Keywords: Data Augmentation, Image Classification, Transfer Learning, ResNet18, VGG16

Abstract. Classification of fruit and vegetable types and conditions have important roles in the food and agricultural industries. One of the important roles is to ensure the product quality. In this research, the classification of fruit and vegetable types and conditions was carried out using a deep learning approach using the VGG16 and ResNet18 models. The used dataset consists of 11,920 image data. The image data are classified into 20 classes which show the type and the condition of fruits and vegetables. Furthermore, model training was carried out using the dataset using transfer learning techniques on the VGG16 and ResNet18 deep learning architecture. Each model training process was done with and without a data augmentation process. The research results show that the ResNet18 model provides higher accuracy than the VGG16 model, namely 96% using augmentation and 95% without augmentation in the classification of fruit and vegetable types. Related to the data augmentation process for each model, the process itself does not affect significantly on model performance since the model performance has already been good.

- **Popular Pre-trained Models**
- ResNet: Skip connections, very deep networks
- VGG: Simple architecture, good for feature extraction
- EfficientNet: Optimal accuracy-efficiency trade-off
- Vision Transformer: Latest state-of-the-art

Practice

Image Classification using transfer learning: Deteksi Jenis Sampah di Sekitar Kampus

Dataset: https://www.kaggle.com/datasets/farzadnekouei/trash-type-image-dataset

Reference

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THANK YOU!